8 Read the passage and answer the questions that follow.

The human heart beats in response to electrical impulses generated by specialised cells in the heart itself. In a healthy heart, these impulses travel through a natural pathway that coordinates each contraction, driving the heart to beat between 60 and 100 times per minute at rest, as shown in Fig. 8.1 below.

In a condition known as complete heart block, this pathway is disrupted. The lower chambers no longer receive reliable electrical signals, causing the heart to beat slowly and erratically, sometimes fewer than 40 times per minute. This is insufficient to sustain blood circulation, and the patient faces the risk of sudden cardiac arrest.

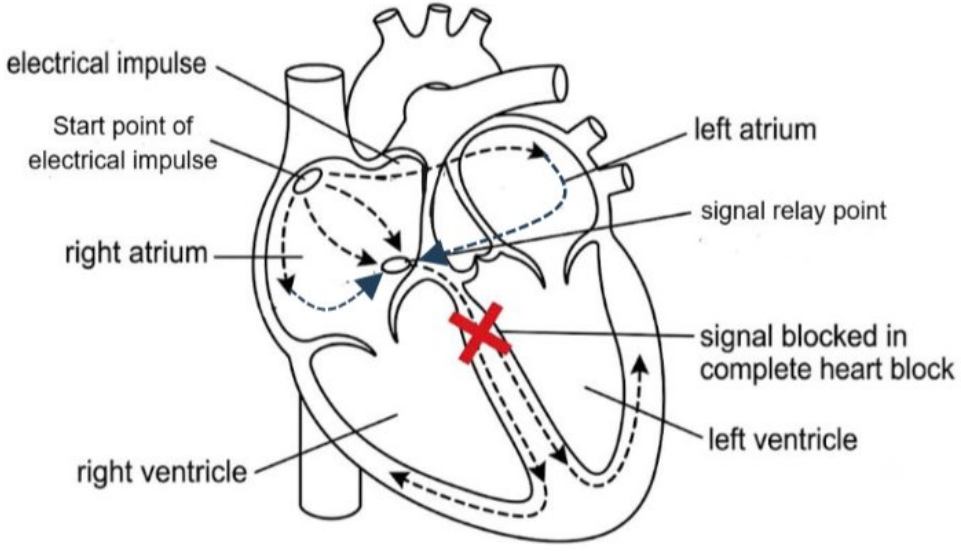


Fig. 8.1

In 1960, electrical engineer Wilson Greatbatch and surgeon William Chardack implanted the world's first fully implantable pacemaker into a human patient.

The device delivered regular electrical pulses to one of the heart's lower pumping chambers (the ventricles) via electrode leads, bypassing the blocked pathway and restoring a steady heart rate of 60 beats per minute. The timing of these pulses was controlled by a simple RC circuit known as a relaxation oscillator.

The pacemaker's timing circuit (RC Relaxation Oscillator) consists of a battery of electromotive force (e.m.f.) ε and a resistor R connected in series, with a capacitor C connected across the output. A voltage-controlled switch S is connected to switch between charging and discharging of the capacitor C.

A simplified circuit diagram is shown in Fig. 8.2 below:

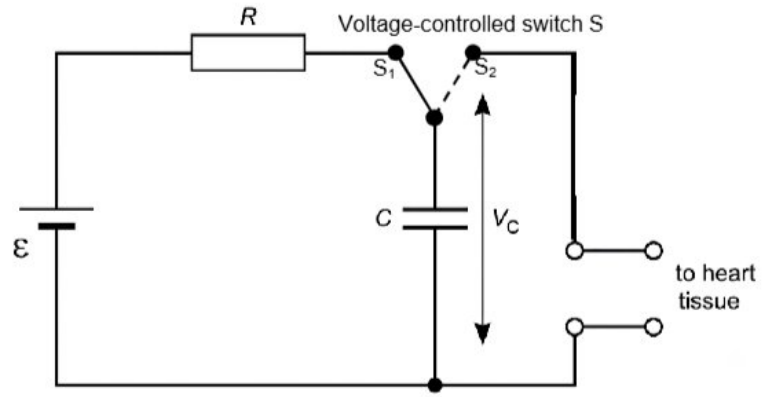


Fig. 8.2

The voltage-controlled switch S responds to the potential difference V_c across the capacitor. It is triggered to S_2 when V_c rises to an upper threshold voltage V_b, and switches to S_1 again when V_c falls to a lower threshold voltage V_s, where $V_s < V_b < \varepsilon$.

When switch S is at S_1, the battery charges the capacitor through the resistor.

When V_c rises to V_b, the switch triggers to S_2 almost instantaneously, and the capacitor discharges rapidly through the heart tissue. This brief discharge delivers a voltage pulse to the heart tissue via output electrodes.

During this discharge, when V_c falls to V_s, the switch returns to S_1 and the capacitor begins recharging. The cycle then repeats, generating a regular train of pulses.

The resulting variation of V_c with time t is shown in Fig. 8.3.

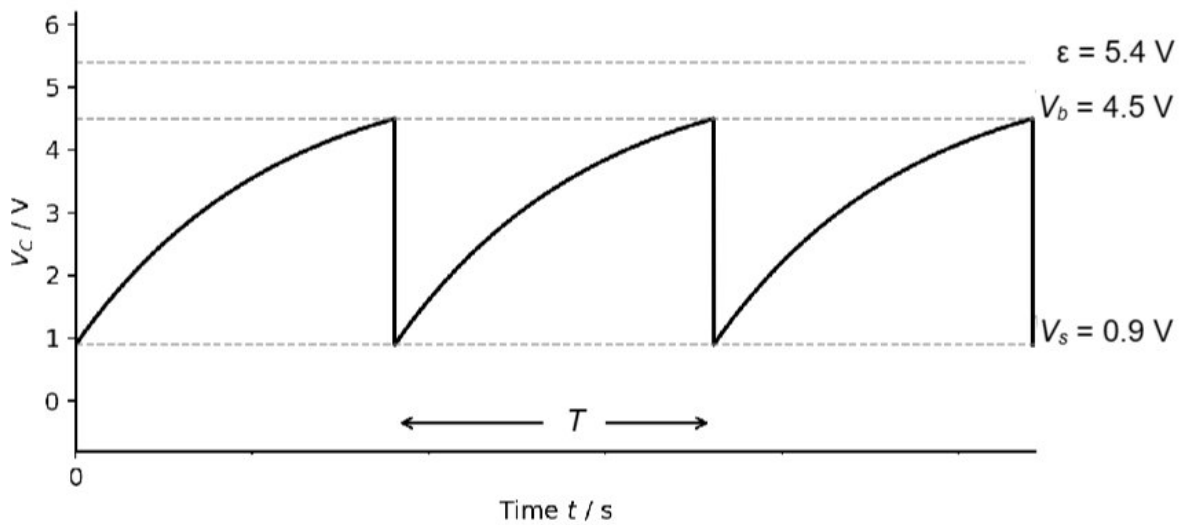


Fig. 8.3

Because the discharge time is very short compared to the charging time, the period T of the oscillation is approximately equal to the time taken for V_c to charge from V_s to V_b. It can be shown that:

$$T \approx RC \ln\left(\frac{\varepsilon - V_S}{\varepsilon - V_b}\right)$$

(a) **(i)** The component values used in the Greatbatch pacemaker are given in the table below.

Component	Value
Timing resistor R	18 kΩ
Timing capacitor C	33 μF

Hence, determine the pacing frequency of the Greatbatch pacemaker and comment whether this value is appropriate for a resting patient. State one limitation of this fixed-rate pacemaker.

Comment: .....

.....

Limitation: .....

.....[4]

(ii) The resistance of the heart tissue is approximately 500 Ω. By calculating the charging and discharging time constants τ, comment on whether the assumption that the discharge is almost instantaneous is justified.

.....

.....[2]

(b) Each time the switch closes, the capacitor discharges from V_b to V_s, delivering a pulse of energy to the heart tissue.

(i) Show that the energy delivered per pulse is 3.21×10^{-4} J. [2]

(ii) Hence, using your answer to (a)(i) and (b)(i), calculate average power delivered to the heart tissue in one charging-discharging cycle.

average power = ... W [2]

(iii) The average power delivered during each discharge pulse is approximately 12.1 mW.

Explain why the average power delivered to the heart tissue in a charging-discharging cycle is much lower than the average power delivered during a discharge pulse.

.....

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.....

.....[2]

(c) The capacity of a battery is the total charge it can deliver, expressed in ampere-hours (Ah), where 1 Ah = 3600 C.

The original mercury-zinc battery used in this pacemaker had a capacity of 1.2 Ah at an e.m.f. of 5.4 V.

(i) Calculate the total electrical energy stored in the mercury-zinc battery.

energy = ... J [2]

(ii) Estimate the expected operational lifespan of the battery in years. Comment on your result.

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.....[2]

(d) Over many years of operation, the capacitance of the timing capacitor will gradually increase due to physical degradation of its dielectric material.

(i) Explain qualitatively why that would result in a decrease in pacing frequency.

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.....[1]

(ii) Suggest, with reasoning, whether a slow decrease in pacing frequency over time would be dangerous for a patient with complete heart block.

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.....[2]

[Total: 19]